



Time Series Analysis

Seminar 8: Long memory and ARFIMA



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Acronyms used in this material

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AIC**: Akaike Information Criterion
- ▣ **AR**: AutoRegressive (model)
- ▣ **ARCH**: AutoRegressive Conditional Heteroskedasticity
- ▣ **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ **ARIMA**: AutoRegressive Integrated Moving Average (model)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **CPI**: Consumer Price Index
- ▣ **DFA**: Detrended Fluctuation Analysis
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **FIGARCH**: Fractionally Integrated GARCH
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GPH**: Geweke–Porter–Hudak (estimator)
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **LW**: Local Whittle (estimator)
- ▣ **MA**: Moving Average
- ▣ **ML**: Machine Learning
- ▣ **OLS**: Ordinary Least Squares
- ▣ **RV**: Realised Volatility
- ▣ **SE**: Standard Error
- ▣ **US**: United States

Today's question and route

- **Question:** does a shock to a series fade at an exponential rate (ARMA) or much more slowly (long memory), and how do we measure the difference?
 - ▶ this seminar comes **before** Lecture 8: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: fractional weights and ARFIMA autocorrelations; R/S and GPH by hand; an ARFIMA forecast; GARCH and FIGARCH weights
 - ▶ Part B: the Nile and its break; Romanian inflation; S&P 500 volatility; BET and EUR/RON
 - ▶ Part C: US inflation and monetary regimes; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	weights of $(1 - L)^d$, ACF of ARFIMA(0, d , 0), ranges of d	Solved, Proposed	A1
A3, A4	R/S and H by hand; a GPH slope by hand	Solved, Proposed	A3
A5, A6	an ARFIMA forecast; GARCH and FIGARCH weights	Solved, Proposed	A5
B1, B2	the Nile and its break; Romanian inflation	Solved, Proposed	B1
B3, B4	S&P 500 volatility; BET and EUR/RON	Solved, Proposed	B3
C1, C2	US inflation and regimes; what is wrong in an AI answer?	Proposed	B1, B2

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
Nile flow at Aswan	statsmodels (nile)	yearly	1871–1970
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2005–2026
CPI, United States	FRED (CPIAUCSL)	monthly, SA	1947–2026
S&P 500, BET	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026

- Romanian inflation: monthly rate $100\Delta \ln P_t$ minus its monthly means (seasonal adjustment), and the 12-month rate $100(\ln P_t - \ln P_{t-12})$
- In the notebook: `gph`, `local_whittle`, `hurst_rs`, `arfima_exact_ml`, `monthly_rv`; no account or key is needed

Prerequisites for Today (1/5): short and long memory

- $\rho(k)$: the autocorrelation at lag k ([Chapter 1](#)); memory = how fast $\rho(k)$ goes to zero
- **Short memory** (ARMA, [Chapter 2](#)): $|\rho(k)| \leq Cr^k$, $0 < r < 1$; the autocorrelations are summable
 - ▶ $C > 0$: a constant; r : the decay rate per lag; AR(1): $\rho(k) = \phi^k$
- **Long memory**: $\rho(k) \sim Ck^{2d-1}$, $0 < d < 1/2$: hyperbolic decay, $\sum_k \rho(k) = \infty$
 - ▶ d : the memory parameter; $\sim$: the ratio of the two sides tends to 1 as $k \rightarrow \infty$
 - ▶ equivalently, the spectral density has a pole at zero: $f(\lambda) \sim G\lambda^{-2d}$ as $\lambda \rightarrow 0$ (λ : frequency in radians; $G > 0$: a constant)
- **Hurst exponent** $H = d + 1/2$: $H = 0.5$ no memory, $H > 0.5$ persistence, $H < 0.5$ anti-persistence
 - ▶ Hurst found $H \approx 0.7$ for the Nile [Hurst \(1951\)](#)

Prerequisites for Today (2/5): fractional differencing and ARFIMA

- Lag operator $Lx_t = x_{t-1}$; the fractional difference $(1 - L)^d$ for real d uses all past values
 - ▶ $(1 - L)^d = \sum_k \pi_k L^k$, with $\pi_0 = 1$, $\pi_k = \pi_{k-1}(k - 1 - d)/k$; π_k : the weight of x_{t-k}
 - ▶ the inverse $(1 - L)^{-d} = \sum_k \psi_k L^k$, $\psi_k = \psi_{k-1}(k - 1 + d)/k$: ψ_k is the effect of a shock after k periods
- **ARFIMA**(p, d, q): $\phi(L)(1 - L)^d(x_t - \mu) = \theta(L)\varepsilon_t$ [Granger and Joyeux \(1980\)](#), [Hosking \(1981\)](#)
 - ▶ $\phi(L)$, $\theta(L)$: the AR and MA polynomials ([Chapter 2](#)); μ : the mean; ε_t : white noise with variance σ^2
- Ranges: invertible for $d > -1/2$; stationary for $d < 1/2$; mean-reverting for $d < 1$; $d = 1$ unit root
 - ▶ if $1/2 \leq d < 3/2$, difference once: $(1 - L)x_t$ is ARFIMA with $d - 1$

Prerequisites for Today (3/5): ARFIMA(0, d , 0) in numbers

- ▣ Autocorrelations by recursion: $\rho(1) = d/(1 - d)$, $\rho(k) = \rho(k - 1)(k - 1 + d)/(k - d)$
- ▣ Variance: $\gamma(0) = \sigma^2 \Gamma(1 - 2d)/\Gamma(1 - d)^2$
 - ▶ $\Gamma(\cdot)$: the gamma function, $\Gamma(n) = (n - 1)!$ for integers; for $d > 0$, $\gamma(0) > \sigma^2$: the memory accumulates past shocks
- ▣ One-step forecast: $\hat{x}_{T+1} = \mu - \sum_{k \geq 1} \pi_k (x_{T+1-k} - \mu)$
 - ▶ x_T : the last observation; since $\pi_k < 0$ for $k \geq 1$ when $0 < d < 1$, every past deviation from μ pushes the forecast in its own direction

Prerequisites for Today (4/5): estimators of d and H

- ▣ **R/S**: in a block of n values, $Y_k = \sum_{i \leq k} (x_i - \bar{x})$, $R = \max Y_k - \min Y_k$, S the standard deviation (divisor n)
 - ▶ $\bar{x}$: the block mean; H = slope of $\log_{10}(R/S)_n$ on $\log_{10} n$; **DFA**: slope of the detrended fluctuation $\log F(n)$
- ▣ **GPH**: OLS of $y_j = \log I(\lambda_j)$ on $X_j = -\log(4 \sin^2(\lambda_j/2))$, $j = 1, \dots, m$; slope = $\hat{d}$, $SE = \pi/\sqrt{24m}$
 - ▶ $I(\lambda_j)$: the periodogram (a noisy estimate of f) at the Fourier frequency $\lambda_j = 2\pi j/T$; T : the number of observations; m : the number of low frequencies used
 - ▶ OLS slope = $\sum(X_j - \bar{X})(y_j - \bar{y}) / \sum(X_j - \bar{X})^2$
- ▣ **Local Whittle** (LW) [Robinson \(1995\)](#): a local likelihood on the same m frequencies, $SE = 1/(2\sqrt{m})$; bandwidth $m = \lfloor T^{0.65} \rfloor$
 - ▶ **Exact ML** [Sowell \(1992\)](#): the full ARFIMA(p, d, q) likelihood; compare models by AIC and BIC

Prerequisites for Today (5/5): volatility and spurious memory

- Returns r_t are almost uncorrelated; $|r_t|$ and realised volatility have long memory [Ding, Granger and Engle \(1993\)](#)
 - ▶ monthly **realised volatility**: $\sqrt{RV_m}$, $RV_m = \sum_{t \in m} r_t^2$, the sum of the squared daily returns of month m ; we model its logarithm
 - ▶ **shuffle test**: permute the days at random; real memory disappears, the distribution stays
- GARCH(1,1) as ARCH(∞): weight $\alpha\beta^{k-1}$ on ε_{t-k}^2 ; FIGARCH [Baillie, Bollerslev and Mikkelsen \(1996\)](#): weights $\approx k^{-1-d}$
 - ▶ ε_{t-k}^2 : the squared shock of k days ago; α, β : the GARCH parameters ([Chapter 5](#)); σ_t^2 : today's conditional variance
 - ▶ with $\phi = \beta = 0$, FIGARCH weights are $\lambda_k = -\pi_k$ of $(1-L)^d$
- **Spurious long memory**: level shifts and rare regime switches give $\hat{d} > 0$ for short-memory series [Diebold and Inoue \(2001\)](#); re-estimate after removing the breaks

A1: fractional weights and the ACF [Solved]

Task

- A series is ARFIMA(0, 0.25, 0) with $\sigma^2 = 1$.
- 1. Compute $\pi_1, \dots, \pi_4$ of $(1 - L)^{0.25}$.
- 2. Compute $\rho(1)$ and $\rho(2)$, and give H and the class of the process.
- 3. Compare $\rho(5)$ with that of an AR(1) with the same $\rho(1)$.
- Report: four weights, two autocorrelations, H , two values of $\rho(5)$.

Solution

- 1. $\pi_1 = -0.25$; $\pi_2 = -0.25 \cdot 0.75/2 = -0.0938$;
 $\pi_3 = -0.0938 \cdot 1.75/3 = -0.0547$;
 $\pi_4 = -0.0547 \cdot 2.75/4 = -0.0376$
- 2. $\rho(1) = 0.25/0.75 = 0.333$;
 $\rho(2) = 0.333 \cdot 1.25/1.75 = 0.238$; $H = 0.75$: stationary, long memory
- 3. ARFIMA: $\rho(5) = 0.151$; AR(1): $0.333^5 = 0.0041$, about 37 times smaller

A2: which range is d in? [Proposed]

Solution discussed in class

Task

- ▣ Two series: (a) ARFIMA(0, 0.6, 0); (b) ARFIMA(0, -0.3, 0). Model: A1.
- ▣ 1. Compute π_1 and π_2 of $(1 - L)^d$ for each.
- ▣ 2. Say whether each series is stationary, invertible and mean-reverting.
- ▣ 3. For (b) compute $\rho(1)$ and $\rho(2)$; for (a) compute $\rho(1)$ of the first difference.
- ▣ Report: four weights, two classifications, three autocorrelations.

A3: R/S by hand and H from three points [Solved]

Task

- ▣ Block: $x = (0.6, 0.4, 0.5, 0.3, -0.4, -0.6, -0.2, -0.5)$. An R/S plot gives $(R/S)_n = 4.1; 9.6; 22.4$ for $n = 16, 64, 256$.
- ▣ 1. Compute $\bar{x}$, the cumulative deviations Y_k , R , S and R/S .
- ▣ 2. Compute H as the slope of $\log_{10}(R/S)_n$ on $\log_{10} n$, and $d = H - 0.5$.
- ▣ Report: R/S , H , d and one sentence.

Solution

- ▣ 1. $\bar{x} = 0.0125$; Y_k : 0.5875, 0.9750, 1.4625, 1.7500, 1.3375, 0.7250, 0.5125, 0; $R = 1.750$, $S = 0.457$, $R/S = 3.83$
- ▣ the signs are grouped (four positive, then four negative): the range is large, as for a persistent series
- ▣ 2. $\log_{10} n$: 1.204, 1.806, 2.408; $\log_{10}(R/S)$: 0.613, 0.982, 1.350; slope $H = 0.61$, $d = 0.11$: moderate persistence

A4: a GPH slope by hand [Proposed]

Solution discussed in class

Task

- ▣ For $m = 4$ frequencies: $X_j = 0.5; 1.6; 2.3; 3.0$ and $\log I(\lambda_j) = 1.2; 1.8; 2.0; 2.5$. Model: A3.
- ▣ 1. Compute $\bar{X}$, $\bar{y}$, $\sum(X_j - \bar{X})(y_j - \bar{y})$ and $\sum(X_j - \bar{X})^2$.
- ▣ 2. Compute $\hat{d}$ and its standard error $\pi/\sqrt{24m}$.
- ▣ 3. Test $H_0: d = 0$ at 5% and comment on the size of m .
- ▣ Report: four sums, $\hat{d}$, its SE, a decision and one sentence.

A5: a one-step ARFIMA forecast [Solved]

Task

- ARFIMA(0, 0.4, 0) with $\mu = 2$; the last four values: $x_T = 3.0$, $x_{T-1} = 2.5$, $x_{T-2} = 2.8$, $x_{T-3} = 1.5$.
- 1. Write the first four weights $-\pi_k$ of $(1 - L)^{0.4}$.
- 2. Compute $\hat{x}_{T+1}$ using these four lags.
- 3. Compare with an AR(1) with the same $\rho(1)$.
- Report: four weights, two forecasts and one sentence.

Solution

- 1. $-\pi_k$: 0.4; 0.12; 0.064; 0.0416
- 2. $\hat{x}_{T+1} =$
 $2 + 0.4(1.0) + 0.12(0.5) + 0.064(0.8) + 0.0416(-0.5) =$
 $2 + 0.4000 + 0.0600 + 0.0512 - 0.0208 = 2.490$
- 3. AR(1): $\phi = \rho(1) = 0.4/0.6 = 0.667$,
 $\hat{x}_{T+1} = 2 + 0.667 \cdot 1.0 = 2.667$; the AR(1) looks only at x_T , ARFIMA spreads the weight over the past (in practice over all T values)

A6: how long does a volatility shock count? [Proposed]

Solution discussed in class

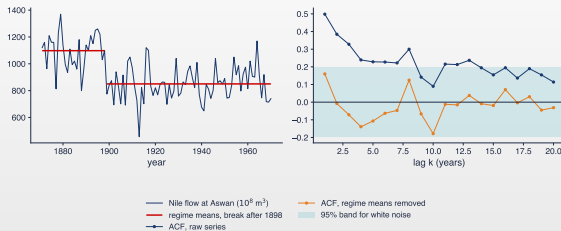
Task

- ▣ GARCH(1,1) with $\alpha = 0.10$, $\beta = 0.88$; FIGARCH(0, d , 0) with $d = 0.4$ (weights $\lambda_k = -\pi_k$). Model: A5.
- ▣ 1. Compute the weight of ε_{t-k}^2 in σ_t^2 for $k = 1, 2, 3$ in both models.
- ▣ 2. Compute both weights at $k = 20$ and $k = 100$; for FIGARCH use $\lambda_k \approx d k^{-1-d} / \Gamma(1-d)$, $\Gamma(0.6) = 1.489$.
- ▣ 3. Explain which model keeps a crisis in today's volatility longer.
- ▣ Report: ten weights and one sentence.

B1: does the Nile have long memory? [Solved]

- ▣ **Question:** is the persistence of the Nile flow long memory, or the effect of the change in 1898?
- ▣ **Inputs:** yearly flow at Aswan, 1871–1970 ($T = 100$)
- ▣ **Tasks**
 1. Plot the series and its ACF up to lag 20.
 2. Estimate H by R/S, d by GPH and local Whittle ($m = \lfloor T^{0.65} \rfloor$) and by exact ML of ARFIMA(0, d , 0).
 3. Subtract the mean of 1871–1898 and the mean of 1899–1970 and repeat step 2.
 4. Interpretation: does the Nile have long memory?
- ▣ **Report:** two ACF values, four estimates before and after, one sentence
- ▣ **Notebook:** section B1

B1: solution [Solved]



- Raw: $\hat{\rho}(1) = 0.50$, $\hat{\rho}(10) = 0.09$; $H_{R/S} = 0.78$; GPH 0.44 (SE 0.15), LW 0.40 (SE 0.11), ML 0.36 (SE 0.07); $m = 19$
- Means 1098 and 850; after removing them: $\hat{\rho}(1) = 0.16$; $H_{R/S} = 0.63$; GPH -0.07 , LW -0.17 , ML 0.05 (SE 0.10)
- Interpretation: for 1871–1970 the evidence of long memory comes from one break; R/S stays above 0.5 because it is biased upwards in short samples

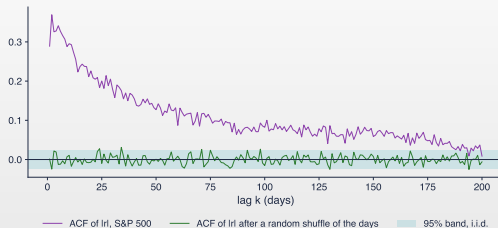
B2: long memory in Romanian inflation [Proposed]

- **Question:** how persistent is Romanian inflation, and does the way we measure inflation change the answer?
- **Inputs:** monthly HICP, 2005–2026: the monthly rate seasonally adjusted with the monthly means, and the 12-month rate; model: B1
- **Tasks**
 1. Estimate d of the monthly rate by GPH and local Whittle with $m = \lfloor T^a \rfloor$, $a = 0.5; 0.65; 0.8$.
 2. Fit ARFIMA(0, d , 0) and ARMA(1,1) by exact ML and compare their BIC.
 3. Estimate d of the 12-month rate by local Whittle and compare the two ACFs.
 4. Interpretation: why does the 12-month rate give d close to 1?
- **Report:** six estimates, two BIC values, one d , two sentences
- **Notebook:** section B2

B3: memory in S&P 500 volatility [Solved]

- **Question:** where is the memory of stock returns: in the returns or in their size?
- **Inputs:** daily log returns of the S&P 500 since 2000 and their monthly realised volatility
- **Tasks**
 1. Estimate d of r_t and of $|r_t|$ by local Whittle and GPH.
 2. Shuffle the days at random and re-estimate d of $|r_t|$; plot both ACFs of $|r_t|$.
 3. Estimate d of the monthly log realised volatility.
 4. Interpretation: what does the shuffle test show?
- **Report:** eight estimates, the chart and one sentence
- **Notebook:** section B3

B3: solution [Solved]



- $T = 6714$ days; r_t : LW -0.01 , GPH 0.02 (SE 0.03 , 0.04); $|r_t|$: LW 0.53 , GPH 0.56
- Shuffled $|r_t|$: LW -0.03 , GPH -0.03 ; ACF of $|r_t|$: 0.289 (lag 1), 0.127 (lag 50), 0.009 (lag 200); log RV (321 months): LW 0.41 , GPH 0.37
- Interpretation: the shuffle keeps every daily value and destroys their order; the memory disappears, so it lives in the clustering of calm and turbulent periods, not in the distribution of returns

B4: BET and EUR/RON [Proposed]

- **Question:** do the BET and the EUR/RON rate show the same memory, and is it stable before and after the 2008 crisis?
- **Inputs:** daily log returns of the BET (since 2000) and of the EUR/RON reference rate (since July 2005); model: B3
- **Tasks**
 1. Estimate d of r_t and $|r_t|$ by local Whittle on the full sample.
 2. Repeat on the subsamples up to 2007 and 2010–2026.
 3. Compare the estimates for r_t with the 95% Monte Carlo band of i.i.d. series of 1900 days.
 4. Interpretation: is the full-sample memory of BET returns real?
- **Report:** twelve estimates, one band and two sentences
- **Notebook:** section B4

C1: US inflation and monetary regimes [Proposed]

- ▣ **Question:** is the long memory of US inflation a property of inflation, or of the changes in monetary policy?
- ▣ **Inputs:** US monthly CPI inflation, annualised, 1947–2026; regimes 1947–1984, 1985–2019, 2020–2026; models: B1, B2
- ▣ **Tasks**
 1. Estimate d by local Whittle on the full sample and in each regime.
 2. Remove the three regime means and re-estimate d .
 3. Propose one more check (a break test, a Markov-switching model, a forecast comparison).
 4. Interpretation: can a constant d describe 80 years of inflation?
- ▣ **Report:** five estimates, one check and a project plan
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant about the long memory of Romanian inflation and of the S&P 500. The answer:
- ▣ (a) Monthly inflation has $d = 0.31$, so its Hurst exponent is $H = d - 0.5 < 0$: it is anti-persistent.
- ▣ (b) The 12-month rate gives $d = 1.00$, so Romanian inflation has a unit root.
- ▣ (c) With $d = 0.31$ the series is stationary and its shocks die out.
- ▣ (d) Use `statsmodels.tsa.arima.model.ARIMA(x, order=(0, 0.31, 0))` to fit the ARFIMA model.
- ▣ (e) S&P 500 returns have $d = -0.01$ and $|r|$ has $d = 0.53$, so returns are predictable.
- ▣ (f) A significant GPH estimate proves long memory.
- ▣ **Tasks**
 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
- ▣ **Report:** a list of six verdicts with one line of justification each

Key takeaways

- ▣ $(1 - L)^d$: weights by the recursion $\pi_k = \pi_{k-1}(k - 1 - d)/k$; ARFIMA(0, d , 0): $\rho(1) = d/(1 - d)$
- ▣ $H = d + 1/2$; R/S and GPH are slopes of log-log regressions
- ▣ Romanian monthly inflation: $d \approx 0.3$; S&P 500: memory in $|r_t|$, not in r_t
- ▣ Breaks can create long memory: the Nile, BET returns, US inflation
- ▣ An AI answer is a draft: check H against d , the library, and the data transformation

After the seminar

- Lecture 8 develops each topic of today
 - ▶ hyperbolic decay, fractional differencing, ARFIMA, the estimators, forecasting, volatility, spurious memory
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: inflation memory and monetary regimes in Romania
- Reading: [Baillie \(1996\)](#); [Graves et al. \(2017\)](#); [Ding, Granger and Engle \(1993\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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